

Assessing the Most Vulnerable Subgroup to Type
II Diabetes Associated with Statin Usage:
Evidence from Electronic Health Record Data
Xinzhou Guo, Waverly Wei, Molei Liu, Tianxi Cai, Chong Wu,
Jingshen Wang

University of Copenhagen

Problem

- Statins are great at treating Coronary artery disease, but may be associated with increased risk of T2D
- Many studies have been conducted, but they either didn't have significant results (didn't use HTE) or were done as randomized controlled trials, which has small sample size and might not be generalizable beyond the study population.

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2. Uncovering which patient subgroup is most vulnerable for developing T2D after taking statins.
3. Assessing the replicability and statistical significance of the most vulnerable subgroup via a bootstrap calibration procedure.

Model Setup

- Logistic Regression model, as outcome is binary.
- Pre-defined subgroups instead of post-hoc indentified subgroups (better when we have prior knowledge to avoid potential bias induced by dataadaptive subgroup identification).

Model Setup

$$\text{logit}\{P(y = 1|z, x)\} = z^T \beta + x^T \gamma, \quad \|\gamma\|_0 \ll p$$

- y is T2D status
- $z \in \mathbb{R}^{p_1 \times n}$ are the subgroup indicator variables
- $x \in \mathbb{R}^{p_2 \times n}$ is the intercept, the subgroup indicators and potential confounders
- $\beta \in \mathbb{R}^{p_1}$ represents the subgroup causal effects on the scale of log odds ratio, i.e. $\beta = (\log \alpha_1, \dots, \log \alpha_{p_1})$ where $\log \alpha_j$ is the log odds ratio of subgroup j .
- It is assumed that $\gamma \in \mathbb{R}^{p_2}$ is a sparse vector with support set M_0 .

Goal 1: Causal identification

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This is done in the appendix but only for 2 and 6 subgroups.

Goal 2 and 3

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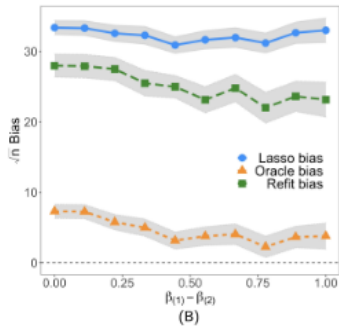
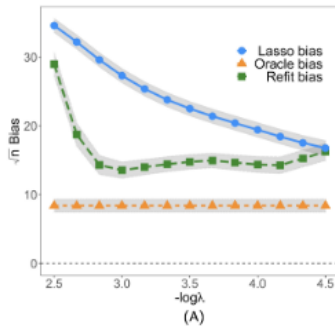
Estimate β_{max}

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2. Winners bias



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1. Accounts for the regularization bias.
2. Accounts for the winners curse bias.
3. Gets the bias-reduced estimate and sharp confidence intervals.

R-split: Step 1

Step 1 (Repeated sample splitting that accounts for the regularization bias) *For $b \leftarrow 1$ to B_1 :* (1)

Randomly split the sample $\{(y_i, x_i, z_i)\}_{i=1}^n$ into two subsamples: a subsample T_1 of size n_1 and a subsample T_2 of size $n_2 = n - n_1$; (2) select a model $\hat{M}_b$ to predict y based on T_1 ; (3) refit the selected model with the data in T_2 to estimate β_b and γ_b via logistic regression:

$$(\hat{\beta}_b^T, \hat{\gamma}_b^T)^T = \arg \min \left\{ \sum_{l \in T_2} \left(y_l \cdot (z_l^T \beta + x_{l, \hat{M}_b}^T \gamma) - \log(1 + \exp(z_l^T \beta + x_{l, \hat{M}_b}^T \gamma)) \right) \right\}.$$

(4) obtain the R-Split estimate: $\tilde{\beta} = \frac{1}{B_1} \sum_{b=1}^{B_1} \hat{\beta}_b$.

R-split: Step 2

Step 2 (Calibrated bootstrap that accounts for the winner's curse bias) For $b \leftarrow 1$ to B_2 : generate bootstrap replicate $\tilde{\beta}^*$ from:

$$\tilde{\beta}^* = \tilde{\beta} + \tilde{\Gamma}_n \cdot \frac{1}{n} \sum_{i=1}^n \begin{pmatrix} z_i \\ x_i \end{pmatrix} \nu_i^*, \quad (2)$$

where $\nu_i^* = u_i \hat{\nu}_i$ is the permuted GLasso residual, $\hat{\nu}_i = y_i - \text{expit}(z_i^\top \hat{\beta}_{\text{GLasso}} + x_i^\top \hat{\gamma}_{\text{GLasso}})$.

Then recalibrate bootstrap statistics via

$$T_b^* = \max_{j \in [p]} (\tilde{\beta}_j^* + \tilde{c}_j(r)) - \tilde{\beta}_{\max}, \quad \tilde{c}_j(r) = (1 - n^{r-0.5})(\tilde{\beta}_{\max} - \tilde{\beta}_j), \text{ where } r \in (0, 0.5).$$

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$$\tilde{\Gamma}_n = \frac{1}{B_1} \sum_{b=1}^{B_1} \mathbf{I}_z \left(\frac{1}{n_1} \sum_{i \in T_{2,b}} f_{i,b} \begin{pmatrix} z_i \\ x_{i, \hat{\mathbb{M}}_b} \end{pmatrix} (z_i^\top, x_{i, \hat{\mathbb{M}}_b}^\top)^\top \right)^{-1} \mathbf{I}_{\hat{\mathbb{M}}_b},$$

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$$f_{i,b} = \text{expit}'(z_i^\top \hat{\beta}_b + x_{i, \hat{M}_b}^\top \hat{\gamma}_b)$$

R-split: Step 2 - Intuition

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Just as $\tilde{\beta}_{\max}$ is a biased estimator of $\beta_{\max}$, the simple bootstrap statistic $\max_{j \in [p_1]} \tilde{\beta}_j^*$ is not centered at $\tilde{\beta}_{\max}$.

R-split: Step 3

Step 3 (Bias-reduced $\tilde{\beta}_{\max}$ and sharp confidence interval) *The level- α two-sided confidence interval for $\beta_{\max}$ is $[\tilde{\beta}_{\max} - Q_{T_b^*}(\alpha/2), \tilde{\beta}_{\max} + Q_{T_b^*}(\alpha/2))$, and a bias-reduced estimate for $\beta_{\max}$ is $\tilde{\beta}_{\max} - \frac{1}{B_2} \sum_{b=1}^{B_2} T_b^*$.*

Choosing tuning parameter r

Step 1 Denote $R = \{r_1, \dots, r_m\}$ as a set of candidate tuning parameters. Randomly split the sample into v equal-sized subsamples.

Step 2 For $l \leftarrow 1$ to m :

For $j \leftarrow 1$ to v :

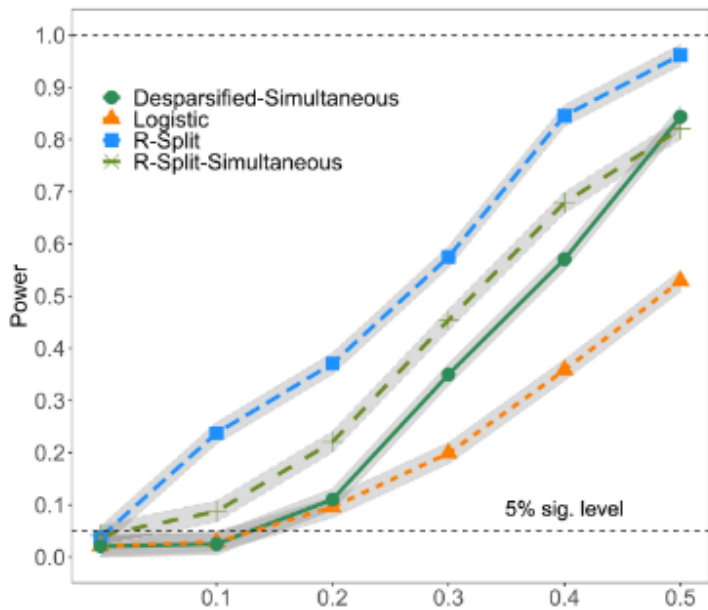
(a) Use subsample j as reference data and the rest as training data. Obtain $\tilde{\beta}_{\max, \text{reduced}, j}(r_l)$ on the training data, where r_l is the tuning parameter.

(b) For $i \leftarrow 1$ to k : Obtain R -Split estimate of $\tilde{\beta}_{i,j}$ and its standard error $\tilde{\sigma}_{i,j}$ on the reference data; evaluate the accuracy

$$h_{i,j}(r_l) = (\tilde{\beta}_{\max, \text{reduced}, j}(r_l) - \tilde{\beta}_{i,j})^2 - \tilde{\sigma}_{i,j}^2.$$

Step 3 Select the tuning parameter via $\arg \min_{r_l} \{ \min_{i \in [k]} [\sum_{j=1}^{j=v} h_{i,j}(r_l) / v] \}$.

Power comparison



Simulations

Their simulations study is for $p_1 \in \{4, 10\}$, while they only prove that β represents subgroup treatment for $p_1 \in \{2, 6\}$.

Limitation

- Only able to make inference on the most vulnerable subgroup.
- Need pre-defined subgroups. Note that they argue that post-hoc identified groups can introduce bias, so if we have prior knowledge it can be best to pre-define the subgroups.
- Some proofs and examples are for their specific case and not generalized.